

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Contact cleaner, Part Number 3824510, or equivalent
- Lubriplate™ 105, Part Number 3163086, or equivalent

Additional Service Items

- Dial indicator
- Hydraulic press

Preparatory Steps

WARNING

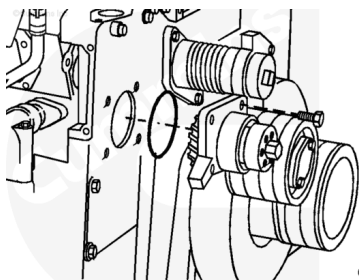
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the belt guards, if equipped. Refer to Procedure 008-001 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-001.html>)
- Remove the charging system alternator drive belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/57/57-013-005-tr.html>)
- Remove the charging system alternator drive pulley. Refer to Procedure 009-010 in Section 9. (</qs3/pubsys2/xml/en/procedures/57/57-009-010.html>)

Remove

Remove the four capscrews and the charging system alternator drive.

Remove and discard the o-ring.

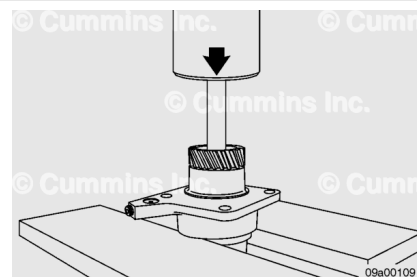


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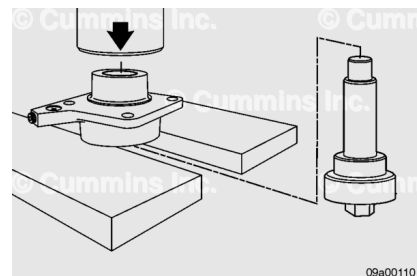
Disassemble

Press the shaft out of the gear and rear thrust plate.

Remove the shaft and hub assembly from the housing.

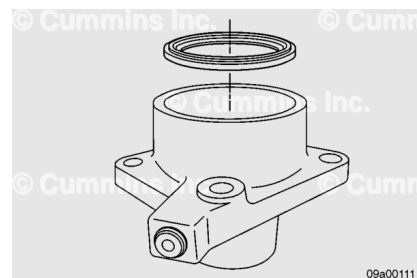


Press the shaft out of the hub and front thrust plate.

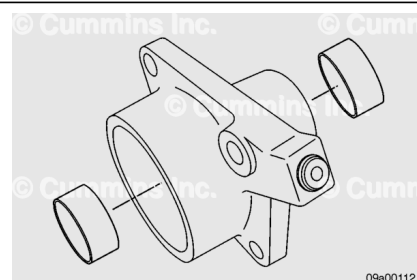


Remove the oil seal from the housing.

Discard the seal.



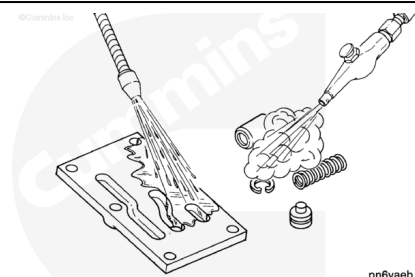
Press the bushings from the front and rear of the housing.



Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

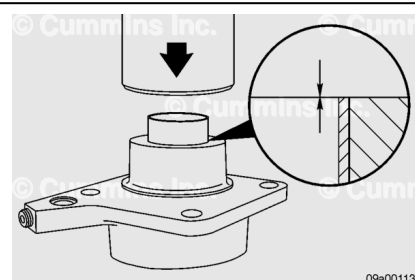
Clean the parts with contact cleaner, Part Number 3824510, or equivalent.

Dry the parts with compressed air.

Assemble

⚠ CAUTION ⚠

Do not use excessive force when pressing components together. Damage to the components can result.

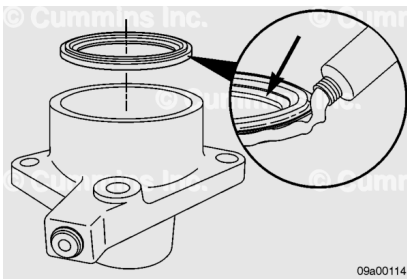


Press a bushing into the front of the housing. Be sure the bushing is flush with the edge of the housing.

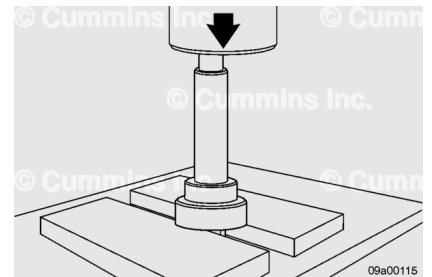
Press a bushing into the rear of the housing. Be sure the bushing is flush with the edge of the housing.

Coat the oil seal with lithium grease (G2-LI or equivalent).

Press the seal into the front of the housing. Be sure the seal is flush with the edge of the housing.



Press the shaft into the hub until the shoulder of the shaft contacts the hub.

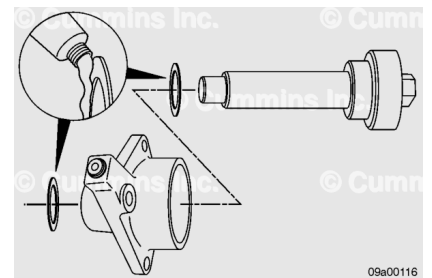


Press the thrust plate onto the shaft.

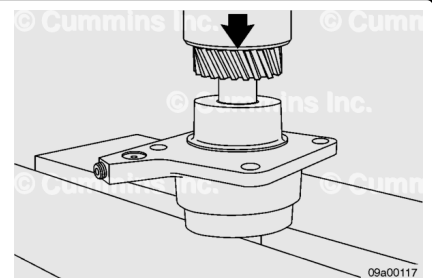
Insert the shaft assembly into the front of the housing.

Press a second thrust plate onto the shaft. Be sure the second thrust plate does **not** ride on the shoulder of the shaft after installation.

Coat both thrust plates with lithium grease (G2-LI or equivalent).



Press or shrink-fit the gear onto the shaft.



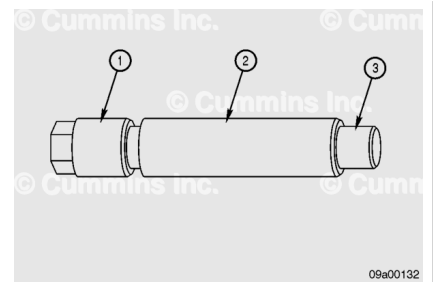
Inspect for Reuse

Clean and inspect the charging system alternator drive shaft for reuse.

Measure the shaft outside diameter.

The charging system alternator drive shaft outside diameter measurement specifications are shown below.

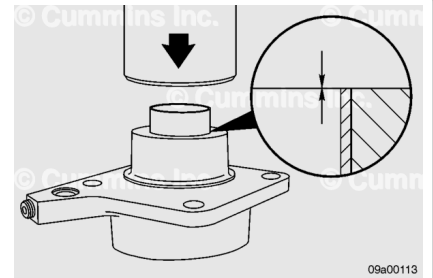
Position	mm		in
1	35.950	MIN	1.4154
	35.965	MAX	1.4159
2	39.959	MIN	1.5732
	39.975	MAX	1.5738
3	30.070	MIN	1.1839
	30.085	MAX	1.1844
If the gear is not within specification, it must be replaced.			



Measure the end clearance.

Charging System Alternator Drive - End Clearance

mm		in
0.10	MIN	0.0040
0.40	MAX	0.0158

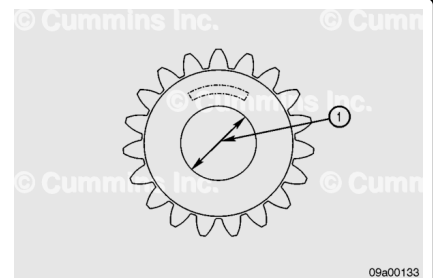


Inspect the teeth of the gear for damage.

Measure the gear inside diameter (1).

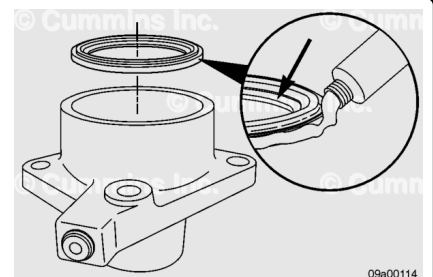
Charging System Alternator Drive - Inside Diameter

mm		in
30.00	MIN	1.1811
30.021	MAX	1.1819



If the gear is **not** within specification, it **must** be replaced.

Inspect the shaft, bushings, and thrust plates for visible signs of wear.



Install

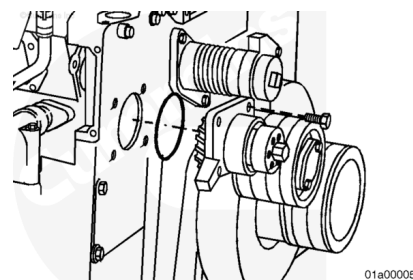
Install the o-ring and check to make sure the o-ring is **not** twisted.

Lubricate the o-ring and the charging system alternator drive bore in the front gear cover.

Install the charging system alternator drive and mounting capscrews into the front gear cover.

Tighten the capscrews.

Torque Value: 113 n•m [83 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the charging system alternator drive pulley. Refer to Procedure 009-010 in Section 9. (</qs3/pubsys2/xml/en/procedures/57/57-009-010.html>)
- Install the charging system alternator drive belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/57/57-013-005-tr.html>)
- Install the belt guards, if equipped. Refer to Procedure 008-001 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-001.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for proper operation.